Delivery Schedule

Name of Bidder:

Bidder's Reference Number:

{Note to Bidders: If the delivery period offered, or any other details, differ from the requirements below, this should be stated in your tender}.

Supply, Installation and Commissioning of Laboratory Information Management Systems at Hwange Power Station

	Description of Goods	OUM	**Qty**	Delivery Date Required by Procuring Entity and applicable INCOTERM	Bidder's offered Delivery period
		OUM	**Qty**	*DDP Hwange*	
1	Supply, Installation and Commissioning of Laboratory Information Management Systems at Hwange Power Station	**Work**	**1**		

PLACE OF DELIVERY

Zimbabwe Power Company
Hwange Power Station
Old Victoria Falls Road
Hwange.

The delivery period required is measured from the date of the signing of the Contract between the Procuring Entity and the Bidder.

Technical Specification and Compliance Sheet

Name of Bidder:

Bidder's Reference Number:

The Goods and Related Services must comply with following Technical Specifications and Standards:

[Columns a and b are completed by the Procuring Entity. Column c must be completed by the Bidder to indicate the full specification of the items offered and their compliance with the specification required (in Column b)]

Supply, Installation and Commissioning of Laboratory Information Management Systems at Hwange Power Station

A	*b*	c
Item No	***Item description and full technical Specification required (including applicable standards)***	***Item description and full technical Specification (Bidder's Offer)***
1.	**SUPPLY, INSTALLATION AND COMMISSIONING OF LABORATORY INFORMATION MANAGEMENT SYSTEM AT HWANGE POWER STATION**	
	1.0 FUNCTIONAL REQUIREMENTS The proposed LIMS shall provide the following core functionalities: **1.1 Sample Management** **1.1.1 Sample Registration:** Ability to register samples manually or via barcode scanners including pre-defined templates for routine samples (e.g., "Boiler Feedwater", "4A Coal Feeder", "Unit 2 Transformer Oil"). 1.1.2 Sample Tracking: Real-time tracking of sample status (Received, In-Progress, Approved, Rejected). 1.1.3 Chain of Custody: Electronic audit trail capturing user ID, timestamp, and actions taken on a sample. **1.2 Test Management & Data Entry** 1.2.1 Configurable Test Methods: The system must allow configuration of test methods based on station standards (e.g., ASTM, ISO, site standard procedures, local codes). 1.2.2 Manual Entry: User-friendly interface for manual input of titration results, pH, conductivity, etc. 1.2.3 Instrument Integration (Auto-Capture): (Mandatory) The system must integrate directly with key	

	laboratory instruments (e.g., Bomb Calorimeters, CHN Analyzers, ICP-OES, Ion Chromatographs) via RS232, USB, or TCP/IP to eliminate transcription errors. **1.3 Calculations & Logic** 1.3.1 Auto-Calculation: The system must calculate derived parameters (e.g., Coal Reactivity, Specific Conductivity, Alkalinity ratios, Calorific Value corrections) based on raw inputs. 1.3.2 Limit Checking: Automatic comparison of results against pre-defined specification limits (Target/Action Levels). Example: • If Boiler pH < 9.0, the system shall flag the result in Red. • If Boiler Silica is > 0.27, the system shall generate a blowdown instruction 1.3.3 Rounding Rules: Configurable rounding logic as per station reporting standards. **1.4 Specific Module Requirements** 1.4.1 Water Chemistry Module: Support for pH, Conductivity, Silica, Dissolved Oxygen, Sodium, Iron, Copper, Cation Conductivity Hydrazine, and Ammonia. Trending capabilities for test parameter vs time eg "Cation Conductivity" vs. "Time". 1.4.2 Coal/Combustion Module: Management of Proximate Analysis (Moisture, Ash, Volatile Matter) and Ultimate Analysis (C, H, N, S, O). Calculation of boiler efficiency inputs. Generation of incoming coal acceptance/rejection certificates 1.4.3 Oil & Fuel Module: Tracking of transformer oil quality including Dissolved Gas Analysis (DGA) trends. Diesel and Heavy Fuel Oil specification management. 1.4.4 Gas Module: Tracking of Hydrogen Purity (%) and Hydrogen Dew Point (°C) with trend plotting as per recent operational reports. 1.4.5 Reporting & Dashboards	

	Standard Reports: Shift logs, Daily Chemistry Reports, Monthly Compliance Reports. Certificates of Analysis (COA): Auto-generation of COAs for dispatch to the Station Management or Regulatory Bodies. Management Dashboard: Visual KPIs showing compliance rates (e.g., % time pH within limits).	
	2. TECHNICAL ARCHITECTURE & INTEGRATION **2.1 Architecture** 2.1.1 Architecture: Client-Server or Web-Based (preferred). If web-based, must support secure intranet deployment. 2.1.2 Database: Industry standard Relational Database Management System (RDBMS) e.g., Microsoft SQL Server or Oracle. 2.1.3 Operating System: Compatible with Windows Server 2019/2022 and Windows 10/11 clients. **2.2 Integration (Critical Requirement)** 2.2.1 DCS/Historian Interface: The LIMS must have drivers or API capability to fetch/process data from the plant DCS or Historian (e.g., OSIsoft PI System, Siemens PCS7). 2.2.2 Use Case: Fetching continuous online pH/Conductivity values to correlate with lab grab samples. 2.2.3 ERP Integration: Capability to export data to the station's ERP system for inventory management of chemicals. **2.3 Security & Compliance** 2.3.1 ISO 17025 Compliance: The system must support 21 CFR Part 11 equivalent electronic signature capabilities. 2.3.2 User Access Control: Role-Based Access Control (RBAC) defining privileges for: • Lab Technician (Data Entry) • Lab Supervisor (Approval) • Station Chemist/Head Chemical Services (Admin/Configuration) • Plant Management (Read-only/Reports).	

	2.3.3 Audit Trail: Immutable audit trails for any creation, modification, or deletion of data. **2.4 Integration (Critical Requirement)** 2.4.1 DCS/Historian Interface: The LIMS must have drivers or API capability to fetch/process data from the plant DCS or Historian (e.g., OSIsoft PI System, Siemens PCS7). 2.4.2 Use Case: Fetching continuous online pH/Conductivity values to correlate with lab grab samples. 2.4.3 ERP Integration: Capability to export data to the station's ERP system for inventory management of chemicals. **2.5 Security & Compliance** 2.5.1 ISO 17025 Compliance: The system must support 21 CFR Part 11 equivalent electronic signature capabilities. 2.5.2 User Access Control: Role-Based Access Control (RBAC) defining privileges for: • Lab Technician (Data Entry) • Lab Supervisor (Approval) • Station Chemist/Head Chemical Services (Admin/Configuration) • Plant Management (Read-only/Reports). 2.5.3 Audit Trail: Immutable audit trails for any creation, modification, or deletion of data.	
	3.0 HARDWARE & SUPPLY REQUIREMENTS The Vendor shall supply the following hardware (or specify specifications if Client-provided): **3.1 LIMS Server:** • High-performance rack-mountable server. • Processor: Minimum 2x Intel Xeon Gold (or equivalent). • RAM: Minimum 64 GB DDR4. • Storage: RAID 5 configuration with minimum 4TB usable storage (SSD/SAS hybrid). • Backup: Tape drive or dedicated NAS connectivity. **3.2 Client Workstations (x8 minimum):**	

	3.2.1 Desktop PCs (i7 processor or higher, 16GB RAM) located in the Chemical Lab, Coal Lab, Foreman Office, Station Chemist Office, Water Treatment Plant Control Room, Head Chemical Services and Plant Management. **3.3 Peripherals:** 3.3.1 Barcode Label Printer (Industrial grade for sample bottles). 3.3.2 Barcode Scanners (Handheld, wireless). 3.3.3 Electronic Demo version of the software showing all functional capabilities (sample management; test management; module management; instrument intergration and workflow management) – Compulsory to be supplied at tendering stage 3.3.4 Network Switch (if dedicated LIMS network segment required).	
	4. INSTALLATION & COMMISSIONING **4.1 Site Preparation** 4.1.1 The Vendor must conduct a site survey to verify network points and power availability. 4.1.2 Installation 4.1.2 Installation of Server OS, Database, and LIMS Application. Configuration of connectivity to more than 25 major laboratory instruments (list to be provided during tender stage). **4.2 Data Migration** 4.2.1 Migration of existing historical data (Excel spreadsheets) into the new LIMS database for trend continuity (minimum 2 years of historical data). **4.3 Commissioning** 4.3.1 Functional Acceptance Testing (FAT) to be conducted at Vendor's premises. 4.3.2 Site Acceptance Testing (SAT) to verify integration and reporting capabilities at Hwange Power Station.	
	5. TRAINING REQUIREMENTS 5.1 The Vendor shall provide comprehensive training for different user groups:	

Training Group	Target Audience	Duration	Content
System Admin	Head Chem, Station Chemist, IT Dept	5 Days +	System Configuration, User management, backup/restore, instrument interface configuration
Power Users	Laboratory Supervisors	3 Days +	Sample registration, result approval, complex reporting, troubleshooting
End Users	Laboratory Technicians	2 Days +	Sample Login, data entry, barcode printing, viewing results
Management	Plant Managers, Engineers	0.5 Days +6	Accessing reports, dashboards and exporting data

5.2 Training materials (User Manuals and Quick Reference Guides) must be provided in hard copy and digital formats

6.0 MAINTENANCE & SUPPORT

6.1 Warranty

6.1.1 The system shall carry a "12-month warranty" from the date of Site Acceptance Test (SAT).

6.1.2 Warranty shall cover software bugs, hardware failures, and patch updates.

6.2 Annual Maintenance Contract (AMC)

6.2.1 After the warranty period, a renewable Annual Maintenance Contract is required for a minimum period of 5 years.

6.2.2 SLA Requirements:

	• Critical Issues (System Down): Response time < 4 hours; Resolution time < 8 hours. • Major Issues (Feature not working): Response time < 8 hours. • Minor Issues (Cosmetic): Response time < 48 hours. • Remote Support: Secure remote diagnostic access must be available. • Updates: Provision for major version upgrades and regulatory patches.	
	7.0 VENDOR QUALIFICATION **7.1 The Vendor must demonstrate:** 7.1.1 Experience in implementing LIMS in Power Generation or Heavy Industry (Minimum 3 reference sites). 7.1.2 Technical capability to integrate with industrial control systems (DCS/SCADA). 7.1.3 Local presence or authorized partner within Zimbabwe or the SADC region for rapid on-site support.	

The detailed technical evaluation will examine the technical specification of the items offered in column c and determine whether this meets the minimum specification in column b. Bidders must complete column c or their tender will be rejected unless supported by technical literature. **Bidders are required to include technical literature to positively support the details/ write in full their specifications provided in column c. Simple statements such as e.g. Comply or Complaint will not be accepted unless supported by relevant technical literature.**